MPPT and Output Voltage Control of Photovoltaic Systems Using a Single-Switch DC-DC Converter

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***Abstract*—This paper presents the simultaneous Maximum Power Point Tracking (MPPT) and output voltage regulation of a stand-alone Photovoltaic-battery-load power system, based on a Single-Switch DC-DC converter. In the literature, the well-known two-stage buck converter cascaded with a buck-boost one is mentioned as the most suitable non-isolated DC-DC converter for such a Photovoltaic (PV) system. Based on an unified approach to develop Single-Stage power converters, we propose to use an equivalent Single Switch Converter (SSC) for the studied PV system, thus increasing its reliability and decreasing the overall system cost, thanks to the reduced number of switches. In the same spirit, the MPPT algorithm has been chosen among the most used ones. The Perturb and Observe (P&O) algorithm appears as the best compromise, performing good tracking factor, simple implementation and satisfactory accuracy. The resulting PV system has been modeled and simulated; the results confirm the effectiveness and the robustness of the simultaneous MPPT and output voltage regulation.**

***Index Terms*—DC-DC power conversion, single-switch, photovoltaic, Maximum Power Point Tracking (MPPT), synchronous switch**

I. Introduction

In the past decades, due to the increasing environmental concerns, many research works have targeted the use of Renewable Energy Sources as an alternative to the fossil fuels. Within this framework, the use of Photovoltaic (PV) systems has seen a remarkable growth and a significant penetration in many fields of application. Nevertheless, in the recent years, the governments have decided to significantly reduce the subsidies for PV systems. In such a context, the plant lifetime is a key element in the estimation of the return of investment [1]. The degradation of the power electronics components of a PV system, especially the switches, contributes significantly to the reduction of its lifetime and reliability. The recent state of the art in PV reliability confirms the fact that usual power electronic converters are the most critical parts in terms of failure rate, lifetime and maintenance cost [2]. Based on reliability studies, the failure rate of a DC-DC power converter is 41%. Semiconductor and soldering joints failures in power switches take up 34% of power electronic system failures [3]. By this way, the reduction of the number of switches seems to be an efficient way to increase lifetime and consequently allow further PV systems penetration. One can notice that reliability must be particularly considered in stand-alone applications.

In this paper, among the various applications of PV systems, we propose the study of a stand-alone Photovoltaic-battery-load power system, based on a non-isolated Single Switch DC-DC Converter (SSC) [4]. In this stand-alone case, a rechargeable battery eliminate the DC-link capacitor voltage stress; moreover, it allows storing energy when the generated solar energy is larger than the one required by the connected load. In the literature, the most suitable non-isolated DC-DC converter in such a stand-alone PV system is the well-known two-stage buck converter cascaded with a buck-boost one [5]. Two switches are used for the DC-DC conversion; each one contributes individually and significantly to reliability decrease and power dissipation. Effectively, in the repeated energy management process, the energy is transmitted from the PV source to the load by flowing from the first stage (buck converter) to the second one (buck-boost converter). Moreover, as mentioned before, a high number of power switches decreases the reliability of the PV system. Additionally, the DC-DC converter cost and its physical size increase with the number of switches, mainly because of the switching pattern generation. Consequently, the reduction of the numbers of power switches also simplifies the switch pattern generation and the associated circuitry.

One possible way to increase reliability and decrease the cost and size of the above mentioned stand-alone PV systems is to apply a unified approach to form a single-stage converter from the cascaded one. An approach based on Synchronous Switches (SS) was first proposed by T. Wu and T. Yu, mainly dedicated to aerospace applications [6]. They have introduced four SS corresponding to the four common node types of two active switches. Here, this design approach is considered when the two switches of the cascade DC-DC converter are turned ON/OFF synchronously. It is applied to the particular MOSFET-based two-stage buck converter cascaded with a buck-boost one. Two constraints must be respected to replace the two switches by a SS: the two switches must share a common node and when both switches are turned ON/OFF synchronously, the converter must have the same behavior as when switches operate individually [6]. Here, by moving toward each other the two switches, the usual cascaded DC-DC converter can be modified to make appear a two-switch common node of Drain-Source connection type, so called "DS: I-II type" by T. Wu and T. Yu [6]. Then, by replacing the two switches by the corresponding SS (here the so called "Inverted IISS:I-IISS", [6]), the Single-Stage Converter (SSC) is obtained.

Let us now consider the control of the PV system. Firstly, it is mandatory to perform Maximum Power Point Tracking (MPPT), aiming at maximizing the extracted energy irrespective of the irradiance conditions [7]. Secondly, the output voltage of the PV system must be also regulated for load feeding

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considerations. The conventional two-stage DC-DC converter can perform MPPT and output voltage regulation simultaneously [8]; when the two switches of the buck and buck-boost converters are controlled synchronously, the duty cycle and the switching frequency are controlled independently to perform simultaneously these two tasks [9]. In the SSC case, the control parameters are the same and are applied to the single switch. Here, among the usual MPPT techniques, the Perturb and Observe (P&0) algorithm appears as the best compromise, performing good tracking factor, simple implementation and satisfactory accuracy [10].

In the next section, the method proposed in [6] is applied to design the SSC from the conventional two stage converter. The major design steps are detailed, in order to simplify the converter circuitry and combine the buck and buck-boost converters to form the SSC, when the two switches are synchronously controlled. In Section III, the SSC operation principle is discussed, mainly based on the energy exchange modes. The SSC control including simultaneous MPPT and output voltage regulation is studied in Section IV. The modeling of the resulting Photovoltaic-battery-load power system is presented in Section V. The PV panel and battery models are briefly discussed. Simulation results are provided and discussed, under variable insolation conditions and variable output voltage references. Finally, in Section V some conclusions are drawn.

II. Single-switch converter (SSC)

A. Topology derivation

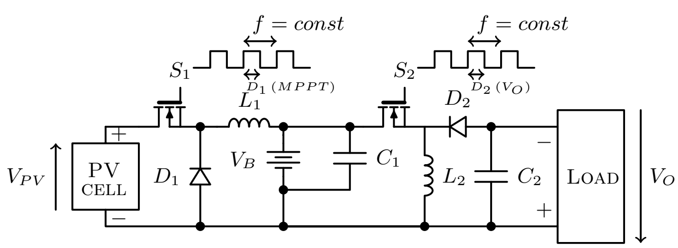

Fig. 1. Two-stage DC-DC converter: the input Buck converter is used for MPPT; the output Buck-Boost converter is used for output voltage regulation.

A common way to carry out simultaneously the MPPT and the output voltage regulation is to use a two stage DC-DC converter as presented in Fig. 1. Each DC-DC stage can be controlled independently in an asynchronous way by its own control signal driving the gate voltage of its switching transistor. The frequency f of the two control signals is usually the same, whereas the duty ratios D_1 and D_2 of two signals are controlled independently, to satisfy both the maximum power extraction and the output voltage regulation respectively. In this case, it is assumed that both converters operate in the continuous conduction mode (CCM).

Another way of driving two different DC-DC stages is by using synchronous mode of operation. In this mode, the two control signals have the same frequency f and the same unique duty ratio D. To control effectively the MPP and the output voltage, two stages have to operate in different modes of functioning: the first one in the Discontinuous Conduction Mode (DCM) whereas the second one in the CCM or vice versa. In this case, the duty ratio D is used for the output voltage regulation

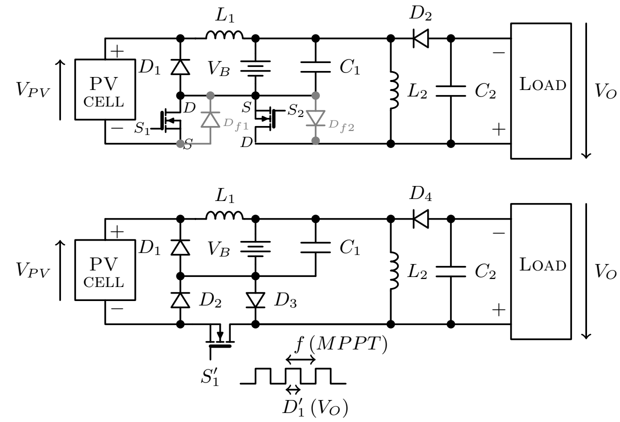

Fig. 2. Two-stage DC-DC converter and its equivalent SSC.

whereas the frequency f of the control signal is used for the MPPT. Thus, a synchronous two stage DC-DC converter has one unique control signal driving the two switching transistors.

The synchronous switching of two DC-DC stages is the starting point of the approach allowing to simplify the initial topology relying on two switching transistors. Indeed, the initial topology presented in Fig. 1 can be restructured as presented at the top of Fig. 2. To have a common node, both switching transistors are moved toward each other. It can be seen that the drain of the switching transistor S_1 is common to the source of the switching transistor S_2. According to [6], two synchronous switches having a common drain-source node and being in the configuration as presented at the top of Fig. 3 can be replaced with an equivalent single-switch topology as presented at the bottom of the same figure. It should be noted that in our initial topology having two switching transistors (see the top of Fig. 2), the diodes D_{f1} and D_{f2} from Fig. 3 cannot be identified. These two diodes are added in the initial topology and are drawn in gray in Fig. 2. By analyzing the initial synchronous topology with the added diodes D_{f1} and D_{f2}, it can be observed that its operation is not altered with these new elements because they are never conducting, due to the inverse polarization supplied either by the PV cell and battery or by the negative load voltage. Thus way, the transformation presented in Fig. 3 from the two-switch synchronous topology to its equivalent single-switch one [6], can be applied to the initial two-stage DC-DC converter, as presented at the bottom of Fig. 2. This single-switch topology is used throughout this work in all further sections as a basis for the MPPT and output voltage regulation.

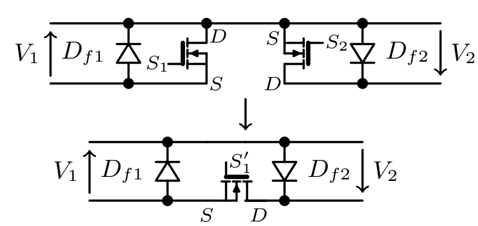

Fig. 3. Synchronous Switches with common Drain-Source node and its equivalent single-switch topology.

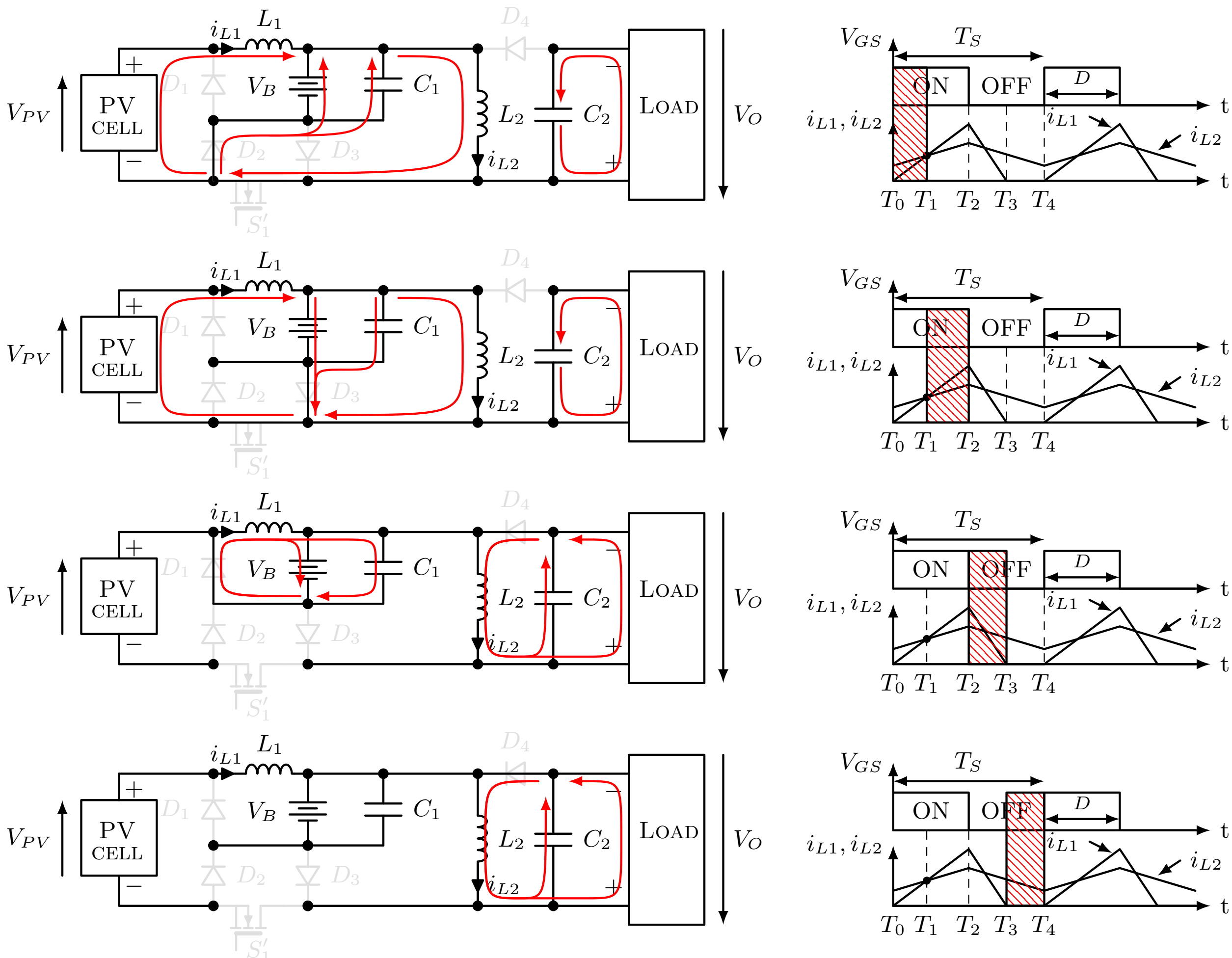

Fig. 4. SSC principle and operation modes.

B. Principle and operating modes of the SSC topology

The operation modes of the single-switch converter topology derived in the previous section are presented in Fig. 4. There are four operation modes, each one is characterized by different energy exchanges between the photovoltaic cell (PV), battery, energy storage elements and the connected load. These energy exchanges are presented in red in Fig. 4.

In the first operating mode, the supplied energy from the PV cell and the battery is first transfered to the inductor L_2 through the inductor L_1, before its further transfer to the connected load in the operating modes 3 and 4. In this operating mode, the connected load is supplied only by the energy stored in the capacitor C_2. This operating mode occurs during the ON state of the switching transistor while $i_{L2} > i_{L1}$.

In the second operating mode, the supplied energy from the PV cell is not only transfered to the inductor L_2, but also used to re-charge the battery and the capacitor C_1. As in the first operating mode, the connected load is supplied with the energy stored in the capacitor C_2. This operating mode occurs during the ON state of the switching transistor when $i_{L2} < i_{L1}$.

The third and forth operating modes occur while the transistor is off. In these operating modes, the connected load is supplied by the energy stored previously in the inductor L_2. The L_2's energy is also used to re-charge the capacitor C_2, whose main function is to supply the connected load during the ON state of the switching transistor. In the third operating mode, the inductor L_1 has still some amount of energy and it uses it to re-charge the battery and the capacitor C_1. Once the total amount of energy from the inductor L_1 is removed, the single-switch converter topology from Fig. 2 moves from third to forth operating mode.

III. SSC CONTROL

A. Perturb and Observe (P&O) technique for MPPT

A PV array under uniform irradiance has a unique MPP, where the array produces maximum output power. This MPP (as well as I-V characteristic of a PV array), changes among the variation of the irradiance level and of the panels temperature [11]. Therefore, to maximize the power produced by a PV system, the MPP should be tracked continuously by a MPPT algorithm. Perturb and Observe ($P\&O$) and INcremental Conductance (INC) MPPT techniques are widely used in commercial products due to its ease of implementation [11], [12].

Generally in all electronic and power electronic converters, control system is necessary to set output voltage on its reference value. In DC-DC converters that are controlled by PWM methods, there are two variables, switching frequency f and duty cycle D, that can be used to control the converter. In the

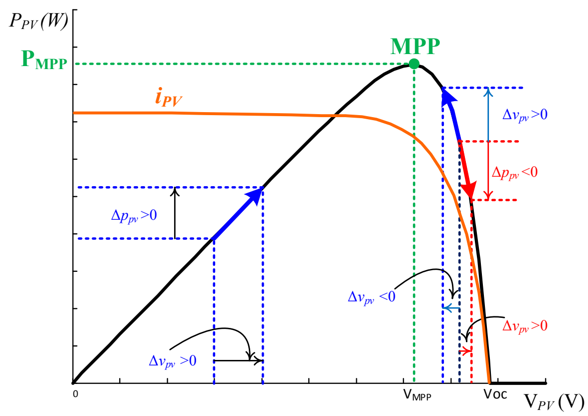

Fig. 5. Principle of the $P\&O$ technique.

proposed SSC, there is just one single switch and two variables (the output voltage V_o and the maximum power of PV P_{PV}) that must be controlled.

Fig. 5 illustrates the principle of the $P\&O$ technique considered in this paper. Here, the operating voltage of the PV (V_{PV}) is perturbed by ΔV_{pv} in a given direction. Consequently, the power drawn from the PV also changes by ΔP_{pv}. If ΔP_{pv} is positive, it means that P_{PV} increases and the operating point has moved toward the MPP. Therefore, the operating voltage must be further perturbed in the same direction. Otherwise, if the P_{PV} decreases, the operating point has moved away from the MPP. Thus, the direction of the ΔV_{pv} must be reversed.

In the SSC topology by increasing (decreasing) the switching frequency f, I_{PV} decreases (increases). Around the MPP of a PV, I_{PV} and V_{PV} have an inverse relationship. Thus, a decrease of the I_{PV} value increases the V_{PV} value and vice versa. Fig. 6 shows the MPPT algorithm based on $P\&O$ with switching frequency (f) as control variable.

B. MPPT and output voltage control

As mentioned previously, the inductor L_2 has to work in the CCM while the inductor L_1 must work in the DCM. In this condition and in steady state operation mode, the relationship between V_B and V_o can be expressed by (1). As the battery's voltage (V_B) can be considered constant, V_O can be controlled by the duty cycle (D).

$$V_o = \frac{D}{1-D} V_B \tag{1}$$

On the other hand, $i_{PV}(t)$ is same as the current through L_1 ($i_{L1}(t)$) when the switch S_1' is on. The average value of the PV current (I_{PV}) which is a function of D, can be expressed by (2).

$$I_{PV} = \frac{I_{L1max} D}{2} \tag{2}$$

While L_1 works in DCM, the I_{L1max} value can be calculated as follows:

$$I_{L1max} = \frac{(V_{PV} - V_B) D}{L_1 f} \tag{3}$$

By replacing (3) in (2) the relationship between I_{PV} and the control variables becomes:

$$I_{PV} = \frac{(V_{PV} - V_B) D^2}{2 L_1 f} \tag{4}$$

As mentioned before, the value of D should be determined by the control of $v_o(t)$. Once the duty ratio D is fixed, the power of PV can be controlled by the switching frequency f by applying the MPPT algorithm depicted in Fig. 6. According to the characteristic (I-V) curve of the PV, V_{PV} increases when I_{PV} decreases ($I_{PV} \downarrow \Rightarrow V_{PV} \uparrow$) and vice versa. Based on the equation (4), to decrease I_{PV}, the switching frequency f must be increased.

To control the output voltage of the SSC topology, two classical controllers are used: the first Proportional Integral (PI) is applied to control $v_o(t)$ and the second PI for $i_{L2}(t)$. To design the PI controllers, the SSC transfer functions are first defined (see equations (5)-(7)), and then the parameters of each PI are determined by using Bode diagrams. Fig. 7 shows a block diagram of the output voltage controlled by the PI controllers.

$$\frac{\tilde{v}_0}{\tilde{d}} = \frac{V_0}{D(1-D)} \frac{1 - \frac{D}{(1-D)^2 R}\frac{L_2}{R} s}{1 + \frac{L_2}{R(1-D)^2} s + \frac{C_2 L_2}{(1-D)^2} s^2} \tag{5}$$

$$\frac{\tilde{v}_0}{\tilde{i}_{L2}} = R \frac{1-D}{1+D} \frac{1 - \frac{D}{(1-D)^2}\frac{L_2}{R} s}{1 + \frac{RC_2}{1+D} s} \tag{6}$$

$$\frac{\tilde{i}_{L2}}{\tilde{d}} = \frac{v_o}{R} \frac{1+D}{D(1-D)^2} \frac{1 + \frac{RC_2}{1+D} s}{1 + \frac{L_2}{R(1-D)^2} s + \frac{C_2 L_2}{(1-D)^2} s^2} \tag{7}$$

A general synoptic of the SSC control is depicted in Fig. 8. As illustrated by this figure, the duty cycle (D) regulates the

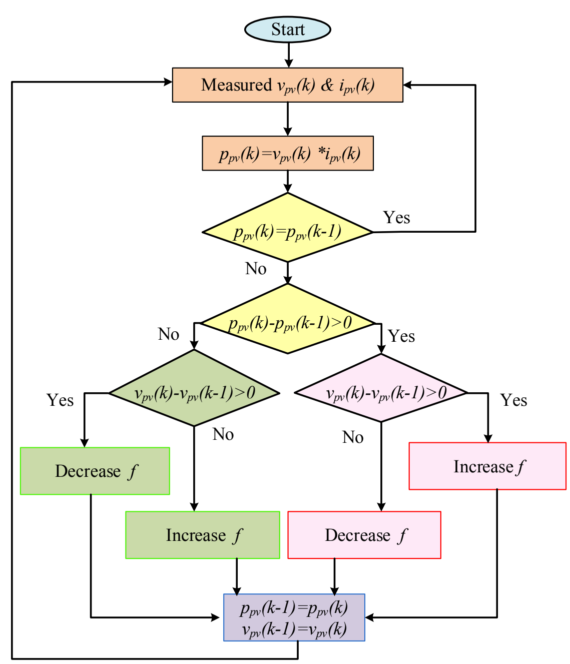

Fig. 6. MPPT algorithm using the $P\&O$ technique.

output voltage and MPPT is done by the switching frequency f.

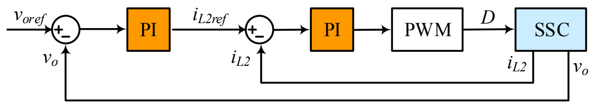

Fig. 7. SSC output voltage control.

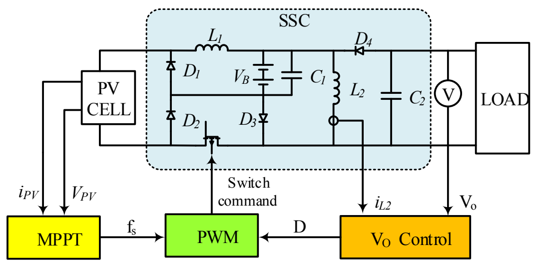

Fig. 8. Synoptic of the SSC control.

IV. System Modeling

A. Photovoltaic cell

In this paper, a model with moderate complexity of a solar cell is used. This model consists of a current source in parallel with a diode and a series resistance, as shown in Fig. 9. The current of the source (I_{ph}) is directly proportional to the light falling on the cell. Increasing sophistication, accuracy and complexity can be introduced to the model by adding some parameters such as the temperature dependence of the diode saturation current [13]–[15]. In Fig. 9, the current I is the difference between the photo current I_{ph} and the normal diode current I_D:

$$I = I_{ph} - I_D = I_{ph} - I_0(e^{\frac{q(V+IR_s)}{mkT_c}} - 1) \tag{8}$$

where m is the idealizing factor, k is the Boltzmann's gas constant, T_c the absolute temperature of the cell, q the electron charge and I_0 is the dark saturation current which depends on the temperature. Generally, a PV module contains PV cells. The PV cells and electrical connectors are encapsulated and covered with various materials to protect them from the environment and form a PV panel.

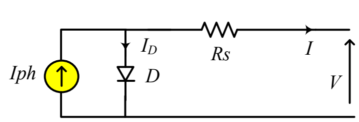

Fig. 9. Equivalent circuit of a PV cell.

B. Battery

A commonly used battery model proposed in [16]–[18] is used in this study. This model is depicted in Fig. 10. Equation (9) expresses the relationship of the battery voltage [16], [18]:

$$V_{bat} = V_{emf} - I_{bat} * (R_1 + \frac{R_2}{(1 + C_b * R_2 * s)}) \tag{9}$$

where V_{bat} is the terminal voltage, V_{emf} is the open-circuit voltage, I_{bat} is the charge/discharge current, R_1 the internal resistance and a parallel RC circuit that illustrates charge transfer and the diffusion between the electrodes and the electrolyte [7], [18].

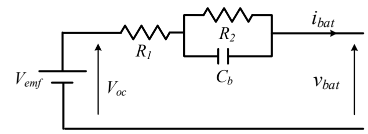

Fig. 10. Equivalent circuit of a battery model.

It is noticeable that the capacitance C_b is sufficiently large such that V_{bat} is taken as a constant value during the modeling stage.

C. Simulation results

To confirm validity of proposed MPPT and output voltage regulation method for the SSC (Fig. 2), some simulations were performed in Matlab/Simulink environment. The SymPowerSystem toolbox is used to model electrical elements of the system. The parameters of the SSC and the PV array are given in Table I.

TABLE I
PARAMETERS OF THE SYSTEM.

	Elements	**Value**		**Elements**	**Value**
PV array	I_{SC}	$0.65A$	SSC	L_1	$15\mu H$
	V_{OC}	$21V$		L_2	$100\mu H$
	R_S	$1.5m\Omega$		C_1	$100\mu F$
	R_P	$10^{11}\Omega$		C_2	$22\mu F$
	n	1.3		V_B	$12V$
	K_i	$0.08\frac{mA}{K}$		R	25Ω

The simulation results are presented in Fig. 11. In the time interval $[0ms, 317ms]$, the irradiance level is $1000\frac{W}{m^2}$ and the panel temperature is set to $25°C$. In this condition, according to the PV characteristic curve, the MPP is equal to $21.4W$. To achieve this MPP, the switching frequency (and thus V_{pv}) is regulated by the MPPT algorithm and its value remains around $51kHz$ for this case. Moreover, the output voltage (V_o) is perfectly regulated by its PI controllers and stabilized around its reference value $(15V)$.

At $t = 317ms$ the irradiance level decreases to $750\frac{W}{m^2}$ and the MPP decreases to $15.8W$. Therefore, the switching frequency f is increased by the MPPT algorithm and I_{pv} decreases (see equation (4)); as a result, the PV produces its maximum power in these new conditions. It is noticeable that, the output voltage V_o remains constant at $15V$.

At $t = 410ms$ the output voltage reference is step-up to $18V$ while the PV conditions remain identical as in the previous simulation step. For this case, the output voltage is correctly regulated at $18V$ by the proposed control. Consequently, the f

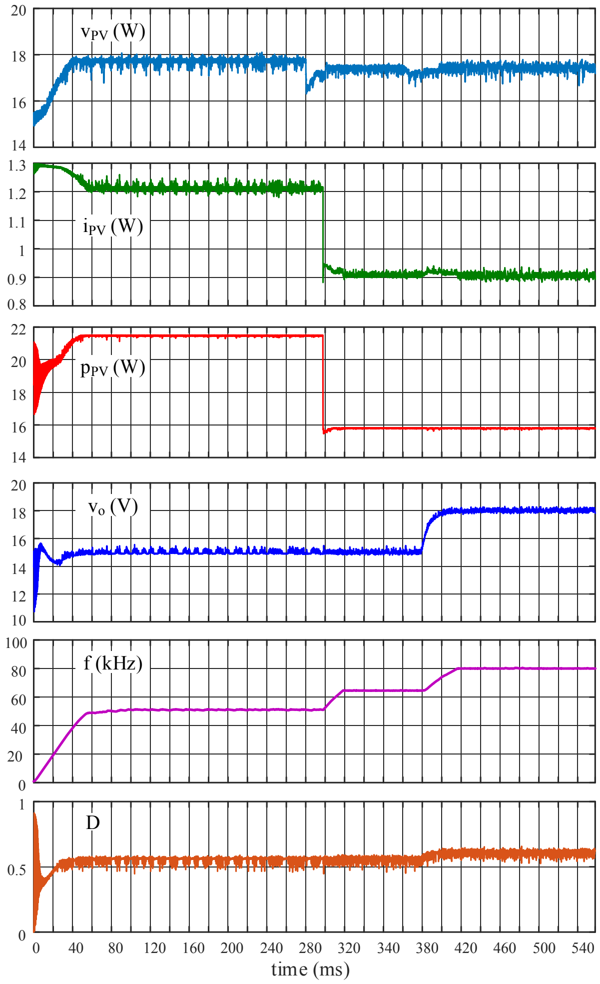

Fig. 11. Simulation results, from up to bottom: V_{pv}, I_{pv}, P_{pv}, V_O, f and D.

value is also increased to keep the PV power at its maximum value.

V. Conclusion

In this paper, to increase reliability and decrease the overall cost of a stand-alone Photovoltaic-battery-load power system, a Single-Switch DC-DC Converter (SSC) instead of a two-stage conventional one, is proposed. It is demonstrated that the conventional two-stage DC-DC converter, composed of a buck converter cascaded with a buck-boost one (see Fig. 2) with the switches operating synchronously, is equivalent to the SSC if a suited control is applied. In steady state condition, one SSC switching period goes through four operation modes, during which the inductors L1 and L2 must work in DCM and CCM, respectively. The SSC control has also been studied. The SS is driven with a PWM control signal having variable switching frequency and duty cycle; the variable switching frequency controls the MPPT whereas the duty cycle control performs output voltage regulation. The Photovoltaic-battery-load power system has been modeled with Matlab environment and simulation results confirm the control effectiveness and the system performances. It can be concluded that the PV system relying on a SSC topology can achieve efficiently the same functionalities as the one using the usual two-stage converter, with increased reliability and lower cost and size. Moreover, the approach presented in this paper could be generalized and applied to any two-stage converter topology. It is particularly interesting when renewable sources are used in stand-alone applications where reliability is a major concern.

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